

## Introduction

A gene's position in a biological network should predict its evolutionary constraint. Genes at the core of signaling cascades, with broad connectivity and pleiotropic tissue expression, should face stronger purifying selection than peripheral effectors with tissue-specific roles because mutations in broadly expressed genes have phenotypic consequences across more biological contexts<sup>1</sup>.

**Melanogenesis offers a test of this hypothesis:** the pathway spans from pleiotropic master regulators (*MITF*, *EDNRB*, *KIT*) to tissue-restricted biosynthetic enzymes (*TYR*, *TYRP1*, *DCT*) acting only in melanocytes.

## Research Question

Do characteristics of a gene's position in the melanogenesis network (e.g. centrality, tissue expression breadth, pathway count) predict its tolerance for genetic variation? Specifically:

- (1) Do breadth and connectivity predict Loss of Function Observed/Expected Upper Bound Fraction (LOEUF) constraint?
- (2) Do any of these factors predict population-specific allelic divergences?

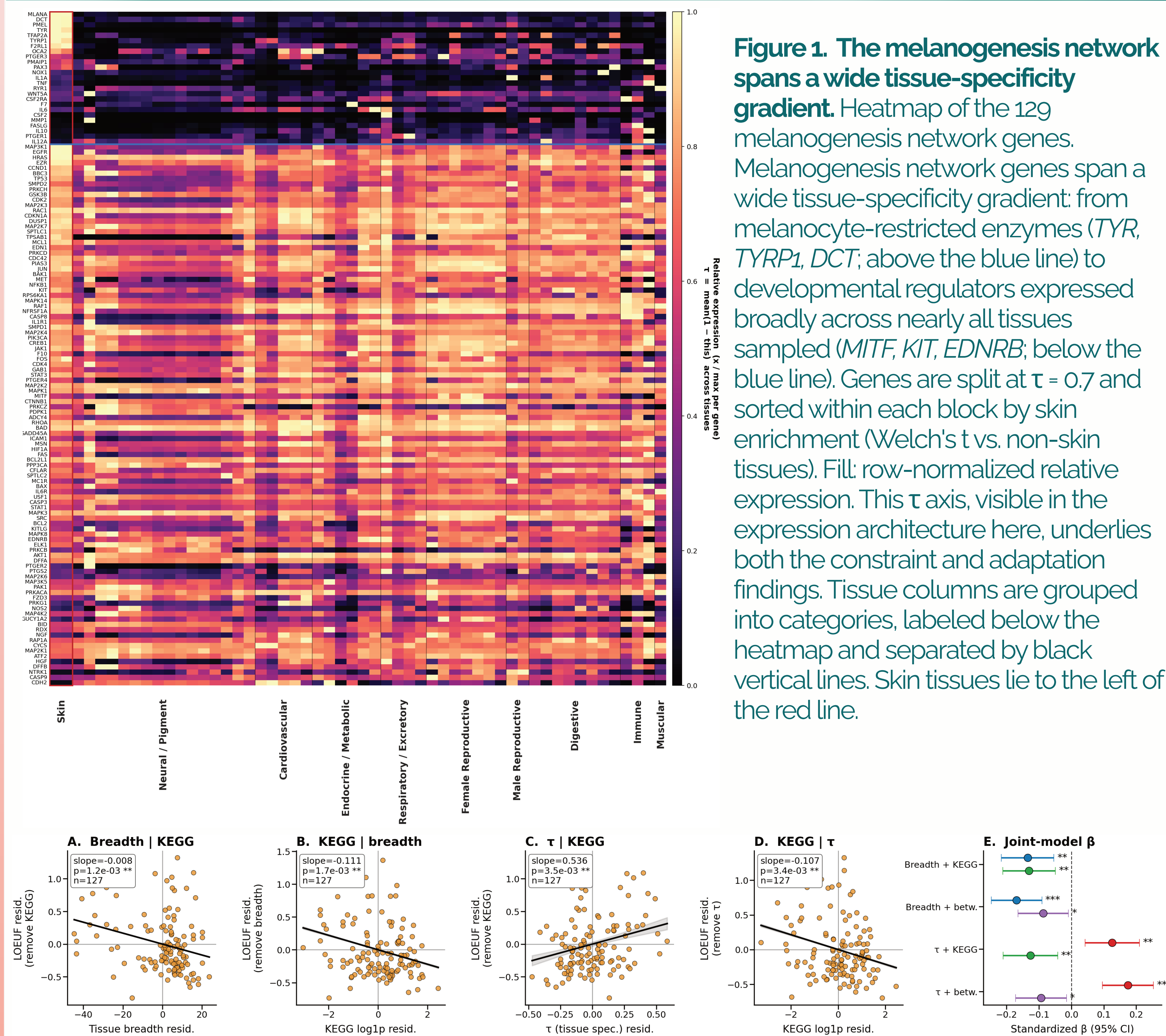
## Methodology

**Gene set:** 129 genes from Raghunath et al. (2015)<sup>3</sup> melanocyte signaling network.

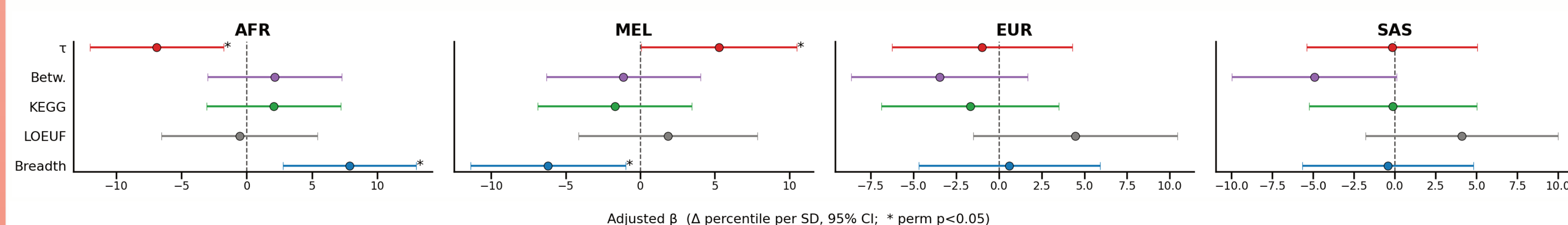
**Constraint metrics:** (1) LOEUF — LoF constraint in modern humans (gnomAD v2.1.1)<sup>2</sup> (2) Tissue breadth — number of tissues with median TPM > 1 (GTEx v8, 54 tissues)<sup>4</sup> (3) Tissue specificity ( $\tau$ )<sup>5</sup> index computed from  $\log_2(\text{TPM}+1)$  across GTEx tissues, where  $\tau = 0$  indicates uniform expression and  $\tau = 1$  indicates tissue-restricted expression. 127 genes were included after QC (biallelic SNPs; GQ  $\geq$  20; site missingness < 10%)

**Population genetics:** Population Branch Statistic (PBS)<sup>6</sup> over 124 regions ( $\pm 10$  kb, excluding 5 region which failed QC-biallelic SNPs; missingness < 10%) in African (n=747; Yoruba, Luhya, Esan, Gambian, Mende, and Mandenka) and Melanesian (n=47) targets, with S. Asian (n=790) and European (n=788) outgroups (gnomAD HGDP+1KGP v3.1.2 + SGDP)<sup>6,7</sup>.

## Findings



**Figure 2. Tissue breadth, specificity ( $\tau$ ) and connectivity (KEGG) independently predict constraint:** Partial regression plots show each predictor's effect on LOEUF after controlling for the other: (A) tissue breadth | KEGG, (B) KEGG | breadth, (C)  $\tau$  | KEGG, (D) KEGG |  $\tau$ . (E) Forest plot of z-scored coefficients ( $\beta \pm 95\%$  CI) from joint OLS models across four bivariate specifications (n = 127). Tissue breadth (A) and KEGG connectivity (B) each independently predict LoF constraint: broadly expressed genes are more constrained, consistent with purifying selection across more biological contexts, and so are genes in more KEGG pathways (breadth  $\beta = -0.135$ ,  $p = 0.001$ ; KEGG  $\beta = -0.131$ ,  $p = 0.002$ ; VIF = 1.16). Tissue-restricted pigment enzymes tolerate loss-of-function; broadly expressed regulators do not. Betweenness centrality predicts LOEUF when paired with breadth alone, but loses significance once KEGG is included (E).



## Discussion

**Pleiotropy, indexed by tissue-expression breadth and network connectivity, predicts LoF constraint in the melanogenesis network. But do not predict patterns in population-specific allele-frequency differentiation.**

(1) Tissue-restricted pigment effectors tolerate LoF; broadly expressed regulators and signaling hubs do not.

(2) Tissue specificity ( $\tau$ ) predicts both constraint and adaptation, but in different ways: broad expression drives purifying selection through pleiotropy; the same tissue-specificity axis determines which populations select which genes.

(3) African PBS is highest at broadly expressed genes; Melanesian PBS at tissue-restricted ones, independent of LoF constraint ( $p = 0.97$ ). Pleiotropy constrains the type of variation that can fix, not which genes diverge between populations.

## Future Work

**Beyond constraint & toward selection targets:** Constraint predicts the space of possible adaptation; tissue specificity determines potential targets of selection. Next steps:

(1) Validate within-network PBS associations in the combined 1KG, SGDP and HGDP genome-wide dataset (larger Melanesian n = 47) to confirm dataset robustness.

(2) Quantify whether pleiotropic constraint on missense variants mirrors LoF constraint and whether African adaptation on broadly expressed genes operates through regulatory rather than coding variation (gnomAD missense Z; eQTL overlap).

(3) Develop a pleiotropic approach to predicting drivers of population-specific selection via GWAS for pigmentation, vitamin D metabolism, and UV response across African and Melanesian populations.

## References

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